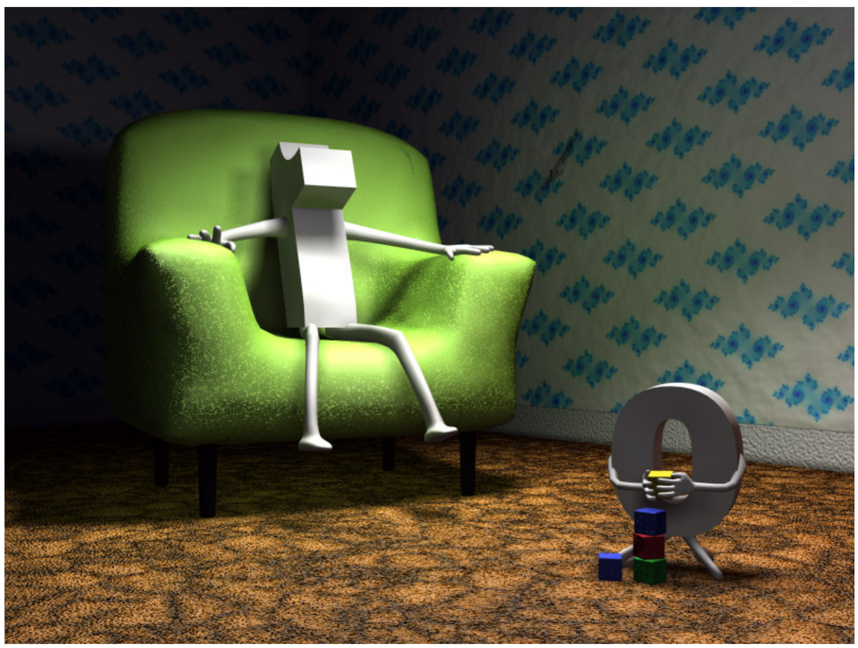
Chapter 1. INTEGERS

1. 1. Induction



Today, we will talk about a property of the natural numbers which is equivalent to Induction.

Well-Ordering Axiom/ Least Integer Principle.
If X is a nonempty subset of $\mathbb{N}$, then there exists $a \in X$ such that $a \leq x$ for all $x \in X$.

Theorem 1.5. *The following principles are equivalent:*

(i) *Well-Ordering*

(ii) *Induction*

(iii) *Strong Induction*

Proof.

(i) $\Rightarrow$ (ii)

Suppose that $P_1, P_2, P_3, \ldots$ are statements satisfying the Basis Step (P_1 is true) and the Inductive Step (P_k is true $\Rightarrow P_{k+1}$ is true, for all $k \geq 1$)

Goal: To show that P_n is true for all $n \geq 1$.

Consider the set $\underline{X} = \{ n \geq 1 \mid p_n \text{ is false} \}$.
We want to prove that $\underline{X} = \phi$. Suppose on the contrary that $\underline{X} \neq \phi$. Then by the Well-Ordering Axiom, $\underline{X}$ has a smallest element, say m.

Notice that $m > 1$ because p_1 is true.
Notice also that $m - 1 \notin \underline{X}$ (because m is the least element of $\underline{X}$). So, p_{m-1} is true and then p_m is true by the inductive step. But this is impossible because p_m is false, since $m \in \underline{X}$, a contradiction. Therefore $\underline{X} = \phi$ and p_n is true for all $n \geq 1$.

(ii) $\Rightarrow$ (iii) ✓

(iii) $\Rightarrow$ (i)

Suppose that X is a nonempty subset of $\mathbb{N}$, and let us show that X contains a least element.

For $n \geq 1$, define the statement

P_n: if $n \in X$, then X has a least element.

Notice that P_1 is true because if $1 \in X$, then 1 is the least element of X.

For $k \geq 1$, assume that $P_1, P_2, \dots, P_k$ are true and let us show that P_{k+1} is true.

Suppose that $k+1 \in X$, and let us prove that X contains a least element.

We consider two cases:

<u>Case 1</u>. $k+1$ is the least element of X

(Nothing to do here)

<u>Case 2</u>. There exists $m \in X$ such that $m < k+1$. Then since $m \in X$ and P_m is true, we have that X contains a least element.

This shows that P_{k+1} is true.

By Strong Induction, we have that P_n is true for all $n \geq 1$. Now, using that $X \neq \emptyset$, there exists $m \in X$. In particular, P_m is true, and since $m \in X$, then X has a least element.

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